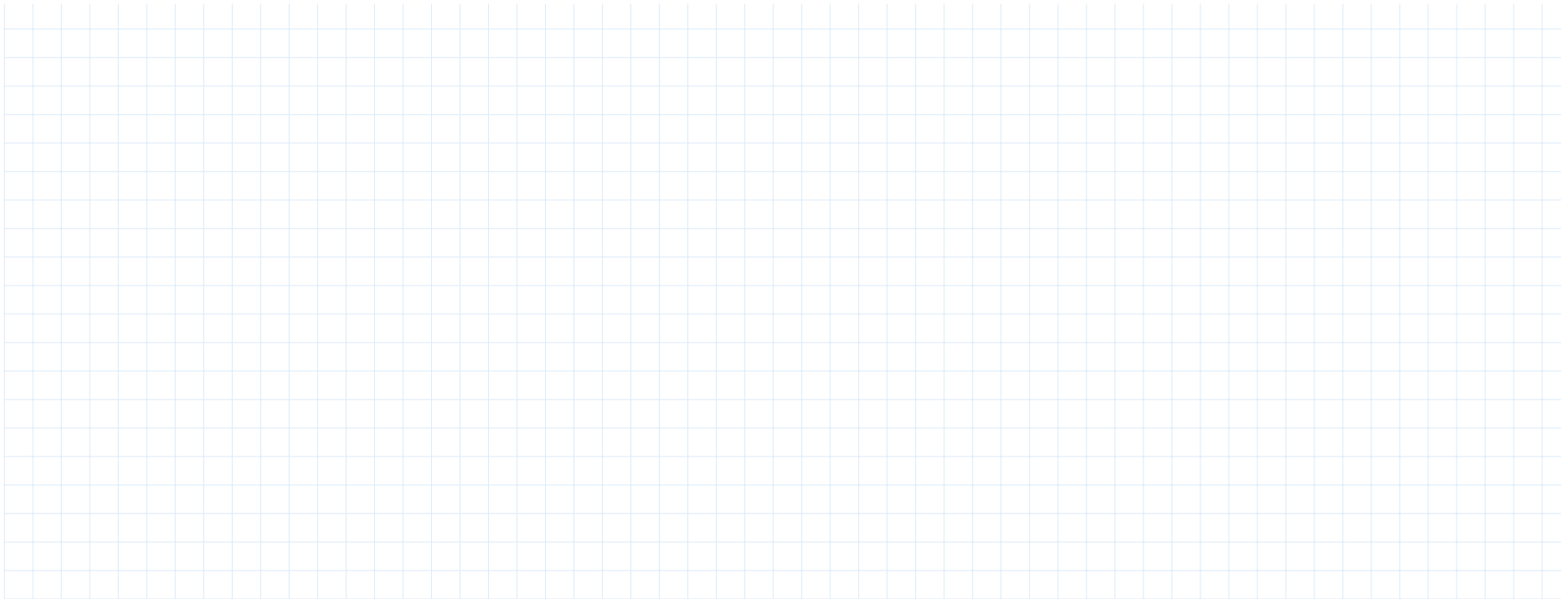


A Logarithm Answers an Exponent Question

For $a > 0$, $a \neq 1$, and $x > 0$,

$$y = \log_a x \iff a^y = x.$$

Evaluate without a calculator: $\log_2 8$, $\log_5 \left(\frac{1}{25}\right)$, and $\log_3 1$. For each answer, write the matching exponential statement.



Translate Between the Two Forms

Rewrite each statement in the other form. Do not approximate.

$$(a) \quad 3^{-4} = \frac{1}{81} \quad (b) \quad e^x = 7 \quad (c) \quad \log_2 64 = 6.$$

Then evaluate $\log_4 64$ and $\log_8 64$ by asking which exponent is needed.



Solve a Logarithmic Equation

Solve each equation. Check the logarithm's argument and base restrictions.

(a) $\log_2(3x - 1) = 3$ (b) $\log_x 16 = 2.$

A logarithm requires a positive argument and a base $a > 0$, $a \neq 1$.



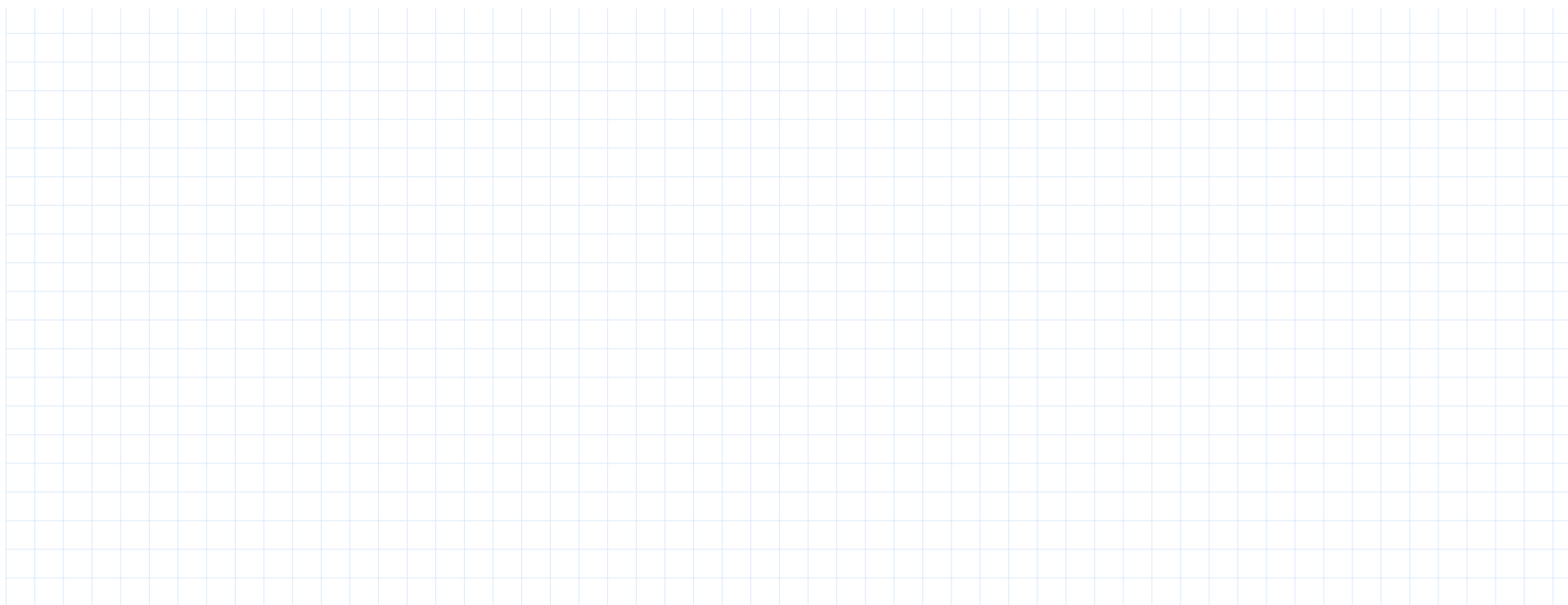
The Basic Logarithmic Graph

The functions $y = \log_2 x$ and $y = 2^x$ are inverses. Use points on 2^x to sketch $y = \log_2 x$. State its domain, range, vertical asymptote, and intercepts. Would $y = \log_{\frac{1}{2}} x$ be increasing or decreasing?



A Shifted Logarithmic Graph

Sketch $f(x) = \log_2(x - 2)$. State the domain, range, vertical asymptote, and intercepts. Explain the shift from $y = \log_2 x$.



Natural and Common Logarithms

Section 4.4

Natural log: $\ln x = \log_e x$. **Common log:** $\log x = \log_{10} x$. Evaluate exactly: $\ln(e^3)$, $\log(0.01)$, and $e^{\ln 5}$. Then state the domain of $h(x) = \ln(4 - x)$. Is $\ln(-2)$ a real number?

